

MATH & PROBABILITY PREREQUISITES PROBLEM LIST

ECO403/503–INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

Optimization

1. Consider the following problems:

- (a) Find *all* functions $f(x) : \mathbb{R} \rightarrow \mathbb{R}$ such that for any $a \in \mathbb{R}$, a is a solution to the maximization problem:

$$\max_{x \in \mathbb{R}} ax - f(x).$$

- (b) Find *all* functions $f(x) : \mathbb{R} \rightarrow \mathbb{R}$ such that for any $a \in \mathbb{R}$, a is a solution to the maximization problem:

$$\max_{x \in \mathbb{R}} af(x) - \frac{x^2}{2}.$$

Hint: Be careful: $f(x)$ doesn't need to be continuous and differentiable!

2. Let X be a compact subset of $\mathbb{R}$; let $f(x)$ and $g(x)$ be continuous functions from X to $\mathbb{R}$. Assume that each function has a unique global maximum point which is x^* for f and x^{**} for g .

- (a) Suppose $g(x) - f(x)$ is a nondecreasing function. Is it true that $x^{**} \geq x^*$?
- (b) Suppose $g(x)/f(x)$ is a nondecreasing function and in addition it is known that f and g are everywhere positive. Is it true that $x^{**} \geq x^*$?
- (c) Consider an arbitrary function $H(u, w)$ which is strictly increasing in the first argument and strictly decreasing in the second argument. It is known that $H(g(x), f(x))$ is nondecreasing in x . Is it true that $x^{**} \geq x^*$?
- (d) If your answer to the previous question is “no”, provide a set of nontrivial conditions on f and g under which it is “yes”.

3. Let $f(x)$ and $g(x)$ be continuous functions from $\mathbb{R}$ to $\mathbb{R}$. It is known that each function has a unique global maximum point which is x^* and x^{**} for f and g respectively. Without loss of generality, let $x^{**} \geq x^*$. Consider the function $h(x) = f(x) + g(x)$.

- (a) Is it true that $h(x)$ attains a maximum on $\mathbb{R}$?
- (b) Suppose that $h(x)$ has a unique global maximum point y . Is it true that $y \in [x^*, x^{**}]$?
- (c) If your answer to the previous question is “no”, provide a set of nontrivial conditions on f and g under which it is “yes”.

4. Consider a function $f(x) : \mathbb{R}^n \rightarrow \mathbb{R}$ and the nonempty compact convex set $X \subset \mathbb{R}^n$. Consider the unconstrained and constrained optimization problems $\max_{x \in \mathbb{R}^n} f(x)$ and $\max_{x \in X} f(x)$. Suppose the problems have unique solutions x^* and x^{**} respectively. It is known that $x^* \notin X$.

- (a) Is it true that x^{**} is on the boundary of X , i.e., at least one of the constraints defining X is binding?
- (b) If no, provide an additional condition on f such that the answer is positive. *Challenge: try to provide the weakest condition.*

5. *Monopoly math: existence and uniqueness.* Consider a standard uniproduct monopoly model with linear pricing. The firm has constant average costs $c \geq 0$, and faces a continuous, decreasing and nonnegative demand function $D(p)$ defined on $[0, \infty)$ such that $D(c) > 0$. The firm maximizes profits.

- (a) Provide a condition under which a solution to the firm's maximization problem exists.
- (b) Assume also that $D(p)$ is continuously differentiable for all p such that $D(p) > 0$. Provide an additional condition under which the solution to the firm's maximization problem is unique. *Hint: one of such conditions concerns the elasticity of demand.*

Probability

1. The joint probability density function of X and Y is given by

$$f(x, y) = c(y^2 - x^2)e^{-y} \quad -y \leq x \leq y, 0 < y < \infty.$$

- (a) Find c
- (b) Find the marginal densities of X and Y .
- (c) Find $E[X]$.

2. The joint probability density function of X and Y is given by

$$f(x, y) = \frac{6}{7}\left(x^2 + \frac{xy}{2}\right) \quad 0 < x < 1, 0 < y < 2.$$

- (a) Verify that it is indeed a joint density function.
- (b) Compute the density function of X .
- (c) Find $\Pr(X > Y)$.
- (d) Find $\Pr(Y > 1/2 | x < 1/2)$.
- (e) Find $E[X]$.
- (f) Find $E[Y]$.

3. The joint probability density function of X and Y is given by

$$f(x, y) = e^{-(x+y)} \quad 0 \leq x < \infty, 0 \leq y < \infty$$

- (a) Find $\Pr(X < Y)$.
- (b) Find $\Pr(X < a)$.

4. The random vector (X, Y) is said to be uniformly distributed over a region R in the plane if, for some constant c , its joint density is

$$f(x, y) = \begin{cases} c & \text{if } (x, y) \in R \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Show that $1/c = \text{area of region } R$.
- (b) Suppose that (X, Y) is distributed over the square centered at $(0,0)$ and with sides of length 2.
 - i. Show that X and Y are independent with each being distributed uniformly over $(-1, 1)$.
 - ii. What is the probability that (X, Y) lies in the circle of radius 1 centered at the origin?

5. Show that $f(x, y) = 1/x$ $0 < y < x < 1$, is a joint density function. Find:

- (a) The marginal density of Y .
- (b) The marginal density of X .
- (c) $E[X]$ and $E[Y]$.

6. The joint density of X and Y is given by

$$f(x, y) = xe^{-(x+y)} \quad x > 0, y > 0.$$

Are X and Y independent? If instead, $f(x, y)$ were given by

$$f(x, y) = 2 \quad 0 < x < y, 0 < y < 1,$$

would X and Y be independent?

7. Let $f(x, y) = 24xy$ $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq x + y \leq 1$.

- (a) Show that $f(x, y)$ is a joint probability density function.
- (b) Find $E[X]$ and $E[Y]$.

8. The joint density function of X and Y is $f(x, y) = x + y$ $0 < x < 1, 0 < y < 1$.

- (a) Are X and Y independent?

(b) Find the density function of X .

(c) Find $\Pr(X + Y < 1)$.

9. The joint density function of X and Y is given by

$$f(x, y) = xe^{-x(y+1)} \quad x > 0, y > 0.$$

(a) Find the conditional density of X , given $Y = y$, and that of Y , given $X = x$.

(b) Find the density function of $Z = XY$.

10. The joint density of X and Y is

$$f(x, y) = c(x^2 - y^2)e^{-x} \quad 0 \leq x < \infty, -x \leq y \leq x.$$

Find the conditional distribution of Y , given $X = x$.

11. If X and Y are jointly continuous with joint density function $f_{X,Y}(x, y)$, show that $X + Y$ is continuous with density function

$$f_{X+Y}(t) = \int_{-\infty}^{\infty} f_{X,Y}(x, t-x) dx.$$

12. Suppose that U_1 and U_2 are independent uniform random variables over $[0, 1]$. What is the distribution of $Z = U_1 + U_2$?

13. Let X_1, X_2, X_3, X_4, X_5 be independent, continuous random variables having common distribution F and density function f , and set

$$I = \Pr(X_1 < X_2 < X_3 < X_4 < X_5).$$

(a) Show that I does not depend on F . *Hint: Write I as a five-dimensional integral and make the change of variables $u_i = F(x_i), i = 1, \dots, 5$.*

(b) Evaluate I .

(c) Give an intuitive explanation for your answer to (b).

14. Let X_1, X_2, X_3 be independent and identically distributed random variables. Compute

(a) $\Pr(X_1 > X_2 | X_1 > X_3)$.

(b) $\Pr(X_1 > X_2 | X_1 < X_3)$.

(c) $\Pr(X_1 > X_2 | X_2 < X_3)$.

(d) $\Pr(X_1 > X_2 | X_2 < X_3)$.

15. Let U denote a random variable uniformly distributed over $[0, 1]$. Compute the conditional distribution of U given that

(a) $U > a$,

(b) $U < a$,

where $0 < a < 1$.